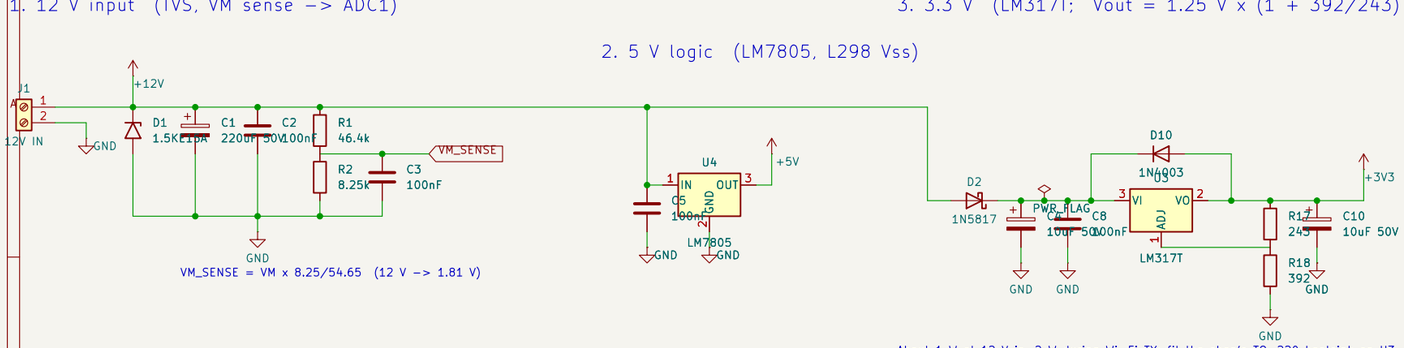
1. 12 V input (TVS, VM sense -> ADC1)

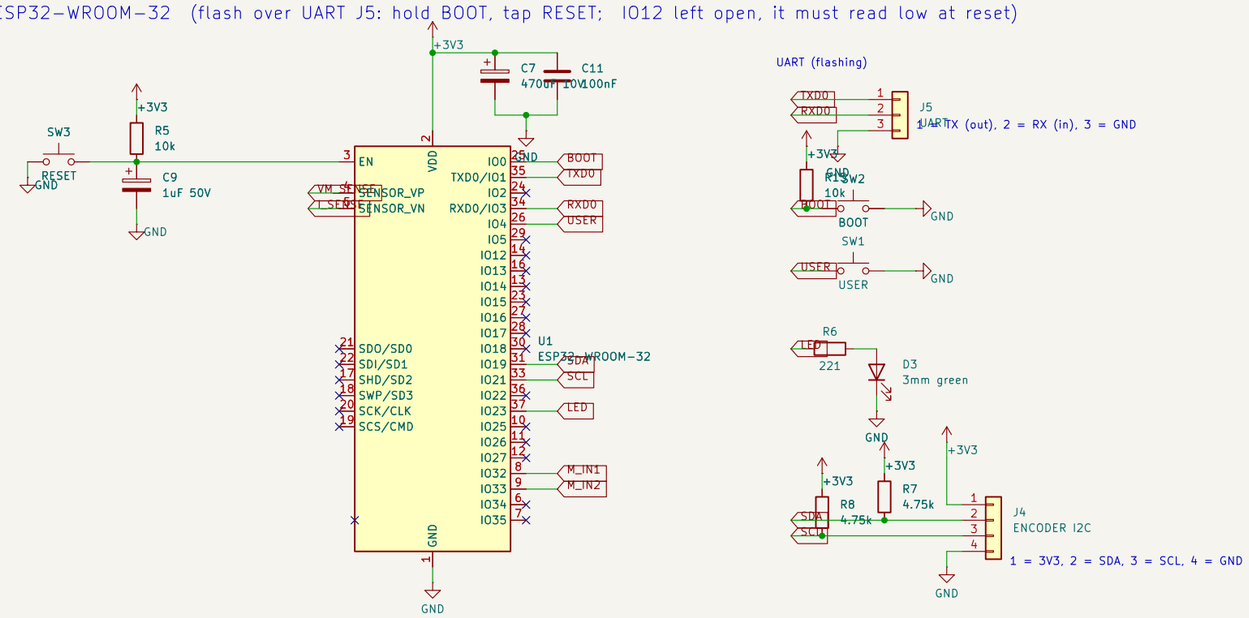
2. 5 V logic (LM7805, L298 Vss)

3. 3.3 V (LM317T; Vout = 1.25 V x (1 + 392/243) = 3.27 V; TO-220 heatsink)

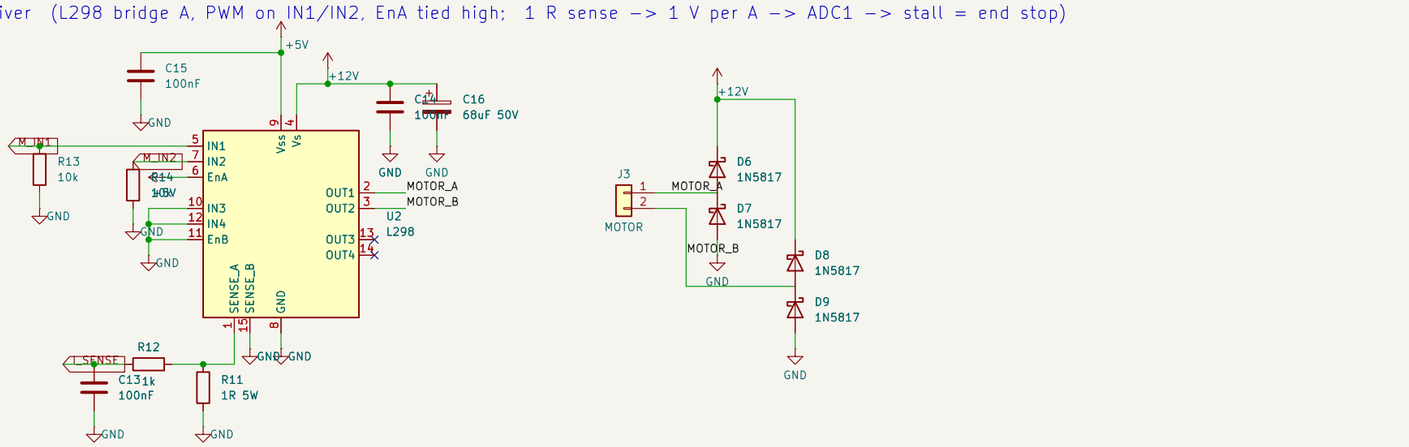
Power flags



4. ESP32-WROOM-32 (flash over UART J5: hold BOOT, tap RESET; IO12 left open, it must read low at reset)



5. Motor driver (L298 bridge A, PWM on IN1/IN2, EnA tied high; 1 R sense -> 1 V per A -> ADC1 -> stall = end stop)



I_SENSE = 1.0 V per amp of motor current (R11 = 1 ohm under bridge A; also flows in slow decay). Running ~0.3-0.8 A, stalled >1.8 A -> firmware stops within 60 ms.

The L298 has no current limit: the stall detection is the only protection, so keep the supply current-limited when bench testing.

Motor is a 6 V rated servo gearmotor on a 12 V bus: firmware caps PWM duty at 50 %. The L298 drops ~2-3 V, so the duty may need raising.

Sheet: /	
File: curtain-drive-revC.kicad_sch	
Title: Curtain Drive Rev C - ESP32-WROOM-32 + L298 curtain motor contro	
Size: A3	Date:
KiCad E.D.A. 10.0.6	Rev:
	Id: 1/1